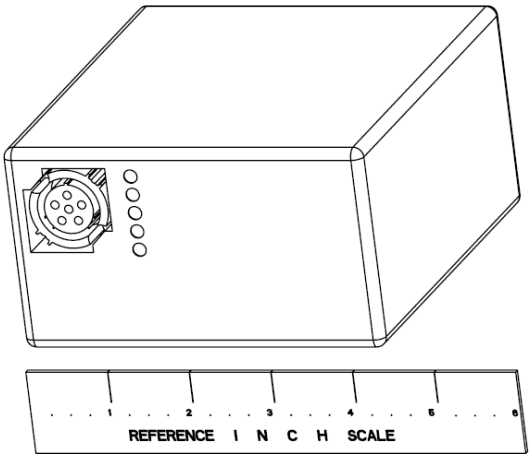
INCH-POUND
MIL-PRF-32271/1C
03 JUNE 2025
SUPERSEDING
MIL-PRF-32271/1B
12 JANUARY 2015

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements.

FIGURE 1. General view, M32271/1 battery shape

For clarity, MIL-PRF-32271 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32271/1C tables are labeled with lower case Roman numerals (e.g. ix).

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	I			II		
Class	2			2		
Features	C	D	E	C	D	E
PIN	M32271/1-12C	M32271/1-12D	M32271/1-12E	M32271/1-22C	M32271/1-22D	M32271/1-22E

Distribution Statement A. Approved for public release: distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

Type	IV		
Class	2		
Features	C	D	E
PIN	M32271/1-42C	M32271/1-42D	M32271/1-42E

Specification requirements: The following requirements of MIL-PRF-32271, identified therein by the phrase “when specified”, are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	**Requirement Paragraph**	**Verification Paragraph**	**Applicability by PIN Suffix (M32271/1-)**								
			12C	**12D**	**12E**	**22C**	**22D**	**22E**	**42C**	**42D**	**42E**
Parallel discharges	Tables II, III, IV, IX, X, XI, & XII		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional material requirements	*3.3.5*	*4.6.1.11*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Parallel cell arrangements[1]	3.4		Yes	Yes	Yes	*No*	*No*	*No*	Yes	Yes	Yes
Connectors	3.4.5	4.6.1.7	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Battery charger connection	*3.4.5.1*	*4.6.1.7.1*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Connection integrity	*3.4.5.2*										
Static	*3.4.5.2*	*4.6.1.7.2*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Dynamic	*3.4.5.2*	*4.6.1.7.3*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Terminal integrity	*3.4.6*										
Socket Strength	*3.4.6.1*	*4.6.1.8.1*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Terminal Strength	*3.4.6.2*	*4.6.1.8.2*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Complete discharge device	3.4.7	4.6.1.9	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State of charge device	3.4.8	4.6.1.10	*No*	Yes	Yes	*No*	Yes	Yes	*No*	Yes	Yes
State of charge data output	3.4.8e	4.6.1.10	*No*	*No*	Yes	*No*	*No*	Yes	*No*	*No*	Yes
Additional design and construction requirements	*3.4.11*	*4.6.1.12*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Cell charging	3.5.3	4.6.2.3	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nail penetration	3.5.4	4.6.2.4	Yes	Yes	Yes	*No*	*No*	*No*	Yes	Yes	Yes
Parallel string charge protection[2]	3.5.6	4.6.2.6	Yes	Yes	Yes	*No*	*No*	*No*	Yes	Yes	Yes
Cell series string short circuit	3.5.7	4.6.2.7	Yes	Yes	Yes	*No*	*No*	*No*	Yes	Yes	Yes
Battery charge protection	3.5.12	4.6.2.12	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Over-current protection	3.5.14.1	4.6.2.14	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Over-temperature protection	3.5.15	4.6.2.15	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Projectile	3.5.16	4.6.2.17	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional safety requirements	*3.5.18*	*4.6.2.18*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Capacity LR	3.6	4.6.4.1.5	*No*	*No*	*No*	Yes	Yes	Yes	*No*	*No*	*No*

[1] Parallel cell arrangements are permitted for Types I and IV batteries described by this specification sheet but are not required.
[2] Only required if parallel cell arrangements are used for Type I or IV batteries.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

TABLE i. *"When specified" requirements* – Continued

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/1-)								
			12C	12D	12E	22C	22D	22E	42C	42D	42E
Capacity LRT	3.6	4.6.4.1.9	*No*	*No*	*No*	Yes	Yes	Yes	*No*	*No*	*No*
Capacity IDC	3.6	4.6.4.1.10	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Surface temperature	3.6h	4.6.4.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Immersion	3.7.8	4.6.5.8	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Watertight integrity	*3.7.9*										
Shallow	*3.7.9*	*4.6.5.9*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Deep	*3.7.9*	*4.6.5.10*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Electromagnetic interference	*3.7.10*	*4.6.5.11*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Electrostatic discharge	*3.7.11*	*4.6.5.12*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
Additional environmental requirements	*3.7.12*	*4.6.5.13*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*
DO NOT CHARGE! marking	3.8.1.2b		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Complete discharge device marking/label	3.8.4		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State of charge marking	3.8.5		*No*	Yes	Yes	*No*	Yes	Yes	*No*	Yes	Yes
Simplified battery label	3.8.7		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional markings	*3.8.8*		*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*	*No*

Dimensions, marking and configuration, all types:

Battery – Drawing A3315878
Battery Connector (3.4.5 & 4.6.1.7) – Drawing SC-C-179492[3]
Mating Connector (3.4.5 & 4.6.1.7) – Part Number SM-D-687888-1, Drawing SM-D-687888
Battery Charger Connector (3.4.5.1 & 4.6.1.7.1) – Not Applicable
Simplified Battery Label (3.8.7) – The energy shall be calculated using the IDC test data.

Battery voltages (3.4.4 & 4.6.1.5), all types:

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):

Type:	Type I		Type II		Type IV	
Connection Mode:	Series	Parallel	Series	Parallel	Series	Parallel
Maximum:	33.00V	16.50V	30.50V	15.25V	33.00V	16.50V
Minimum:	29.80V	14.90V	28.80V	14.40V	29.60V	14.80V

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):

For capacity & initial voltage delay –
10.0 volts minimum for parallel connection
20.0 volts minimum for series connection

[3] Certifications for physical properties of connector material required by SC-C-179492.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

For closed circuit voltage test -
10.0 volts per battery section

Cell closed circuit voltage (3.4.4.3 & 4.6.1.3):

Minimum – 2.0 volts per cell

Maximum weight (3.1 & 4.6.1.6):

Type I - 3.10 lbs. (1,406 g)
Type II – 2.33 lbs. (1,057 g)
Type IV – 2.87 lbs. (1,300 g)

State of charge (3.4.8 & 4.6.1.10): Batteries described by this specification sheet have two independent sections. State of charge circuitry, when required, shall be located in the section connected to sockets 1 and 4. See applicable drawing for additional requirements pertaining to batteries with state of charge data output requirements.

Capacity test requirements (3.6 & 4.6.4), Type I:

Test	Initial voltage delay Time (MAX) to	Volts	Minimum Capacity		Cut-off Volts
			Storage Period		
			1-Week	4-Week	
I	1 second	20.00	5.0 hours		20.00
IP	1 second	10.00	5.0 hours		10.00
IDC	1 second	10.00	40.0 hours		10.00
L	1 second	20.00	3.3 hours		20.00
LP	1 second	10.00	3.3 hours		10.00
H	1 second	20.00	8.3 hours		20.00
HP	1 second	10.00	8.3 hours		10.00
IT	1 second	20.00	4.7 hours	4.5 hours	20.00
ITP	1 second	10.00	4.7 hours	4.5 hours	10.00
LT	1 second	20.00	3.1 hours	3.0 hours	20.00
LTP	1 second	10.00	3.1 hours	3.0 hours	10.00
HT	1 second	20.00	7.7 hours	7.5 hours	20.00
HTP	1 second	10.00	7.7 hours	7.5 hours	10.00

Capacity test requirements (3.6 & 4.6.4), Type II:

Test	Initial voltage delay Time (MAX) to	Volts	Minimum Capacity	Cut-off Volts
I	5 seconds	20.00	3.4 hours	20.00
IP	5 seconds	10.00	3.4 hours	10.00
IDC	5 seconds	10.00	30 hours	10.00
L	60 seconds	20.00	2.2 hours	20.00
LP	60 seconds	10.00	2.2 hours	10.00
H	5 seconds	20.00	6.2 hours	20.00
HP	5 seconds	10.00	6.2 hours	10.00
LR	5 seconds	20.00	50.0 hours	3.00

Continued next page

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

Test	Initial voltage delay Time (MAX)	to Volts	Minimum Capacity Storage Period 1-Week	4-Week	Cut-off Volts
IT	5 seconds	20.00	3.1 hours	3.0 hours	20.00
ITP	5 seconds	10.00	3.1 hours	3.0 hours	10.00
LT	60 seconds	20.00	2.0 hours	1.9 hours	20.00
LTP	60 seconds	10.00	2.0 hours	1.9 hours	10.00
HT	5 seconds	20.00	5.8 hours	5.6 hours	20.00
HTP	5 seconds	10.00	5.8 hours	5.6 hours	10.00
LRT	5 seconds	20.00	46.5 hours	45.0 hours	3.00

Capacity test requirements (3.6 & 4.6.4), Type IV:

Test	Initial voltage delay Time (MAX)	to Volts	Minimum Capacity	Cut-off Volts
I	1 second	20.00	4.5 hours	20.00
IP	1 second	10.00	4.5 hours	10.00
IDC	1 second	10.00	58.0 hours	10.00
L	60 seconds	20.00	7.0 hours	19.50
LP	60 seconds	10.00	7.0 hours	9.75
H	1 second	20.00	8.7 hours	20.00
HP	1 second	10.00	8.7 hours	10.00

			Storage Period 1-Week	4-Week	
IT	1 second	20.00	4.2 hours	4.1 hours	20.00
ITP	1 second	10.00	4.2 hours	4.1 hours	10.00
LT	60 seconds	20.00	6.5 hours	6.4 hours	19.50
LTP	60 seconds	10.00	6.5 hours	6.4 hours	9.75
HT	1 second	20.00	8.1 hours	7.9 hours	20.00
HTP	1 second	10.00	8.1 hours	7.9 hours	10.00

Abuse test pulse discharge capacity requirement, all types (3.5.13 & 4.6.2.13l):

Type I – 9.4 ampere-hours
Type II – 6.2 ampere-hours
Type IV – 8.5 ampere-hours

METHODS OF EXAMINATION AND TEST:

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell to be used in a single series cell string design with 2 amperes constant current, 1 ohm constant resistance, or 4 watts constant power. In the event that parallel cell arrangements are used (Types I and IV only), use a load that produces a current equivalent to 2 amperes divided by the number of parallel cell strings to be used, as a minimum.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load each independent section of the battery with 2 amperes constant current, 5 ohms constant resistance, or 20 watts constant power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

Abuse test, pre-discharge: When tested as specified in 4.6.2.13a, discharge in the series mode with a load of either 2 amperes or 50 watts for: 2.5 hours for Type I batteries; 1.7 hours for Type II batteries; and 2.25 hours for Type IV batteries.

Abuse test, pulse discharge: When tested as specified in 4.6.2.13l, discharge in the series mode with a load of 2.5 amperes for 60 (-0,+1) seconds followed by 1.8 amperes for 240 (-1, +0) seconds cycled continuously to 20.00 volts.

Battery over-current protection: When tested as specified in 4.6.2.14, apply a load to the batteries with the battery sections connected in the series mode and subject to a load of either 4 amperes or 96 watts.

Immersion: When tested as specified in 4.6.5.8, apply a load, with the battery connected in the series mode, of either 0.4 amperes or 4.6 watts during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test[4]	Discharge Rate	Duty Cycle
I	2 amperes	Continuous discharge to zero volts, followed by 2-ampere forced discharge for 10 minutes.
IP	4 amperes	Continuous discharge to zero volts, followed by 4-ampere forced discharge for 10 minutes.
L	2 amperes	Continuous discharge to cut-off voltage.
LP	4 amperes	Continuous discharge to cut-off voltage.
H	2 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
HP	4 amperes	Discharge for 5 minutes at 4 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
LR	145 mA/5.5W	Discharge in series mode at 145 milliamperes to 20.5 volts, then Discharge at 5.5 watts to second cut-off.
IT	2 amperes	Continuous discharge to cut-off voltage.
ITP	4 amperes	Continuous discharge to cut-off voltage.
LT	2 amperes	Continuous discharge to cut-off voltage.
LTP	4 amperes	Continuous discharge to cut-off voltage.
HT	2 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
HTP	4 amperes	Discharge for 5 minutes at 4 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage.
LRT	145 mA/5.5W	Discharge in series mode at 145 milliamperes to 20.5 volts, then Discharge at 5.5 watts to cut-off voltage.
IDC	4.6/6.0/20W	Discharge in parallel for 4.6W for 6 minutes, then 6.0W for 3 minutes, then 20W for 1 minute; repeat cycle until cut-off voltage.

[4] Discharge tests including the letter P require discharge with the two sections of the battery connected in parallel; all other capacity tests require discharge with the two sections connected in series.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This specification covers non-rechargeable lithium batteries of the non-reserve type composed of electrochemical cells using lithium metal in the anode. They are intended for use in military communications and electronics equipment.

Nominal ratings[5]: The following are the nominal ratings for each battery type described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/1-12C, 12D or 12E (Li/MnO_2)**		**M32271/1-22C, 22D or 22E (Li/SO_2)**		**M32271/1-42C, 42D or 42E (Li/CFx-MnO2)**	
Overall Dimensions:	5.00 in. x 2.45 in. x 4.40 in.					
MAX Weight:	3.1 lbs (1.41 Kg)		2.33 lbs (1.06 Kg)		2.87 lbs (1.30 Kg)	
Connection Type:	Series	Parallel	Series	Parallel	Series	Parallel
Voltage Range:	20-33	10-16.5	20-30.5	10-15.25	20-33	10-16.5
Nominal Capacity (I-test) (Ampere-hours): (IDC-test)	10-11 10	20-22 20	6.8-7.2 7.5	13.6-14.4 15	9-10 15	18-20 30
Nominal Energy (Watt-hours):	208-280		160-200		238-400	
MAX Rated Power Output:	50 watts		50 watts		50 watts	
MAX Continuous load Rating:	2 Amps	4 Amps	2 Amps	4 Amps	2 Amps	4 Amps
MAX Pulse Rating:	2.5 Amps	5 Amps	2.5 Amps	5 Amps	2.5 Amps	5 Amps
Instantaneous Trip Rating:	4 Amps	8 Amps	4 Amps	8 Amps	4 Amps	8 Amps
Operating Temperature Range:	-4 to 130ºF (-20 to 55ºC)		-20 to 130ºF (-29 to 55ºC)		-20 to 130ºF (-29 to 55ºC)	
Storage Temperature Range:	-40 to 160ºF (-40 to 71ºC)					
MAX Abusive Temperature (non-operating):	192ºF (89ºC)				192ºF (89ºC)	

Other data:

Military Type Designations. The military type designations that relate to the PINs covered by this specification sheet are as follows: The BA-5390()/U designation has been used for the Type I PINs; the BA-5590()/U designation has been used for the Type II PINs; the BA-5790()/U has been used for the Type IV PINs; and the BA-7590()/U designation was once used for prototypes of the Type III PIN.

All battery types described by this specification sheet have two independent sections with each section designed to provide a nominal 12 volts in use. Sockets 1 and 4 on the connector are for one section; sockets 2 and 5 are the other. Using equipment can connect for either 12- volt or 24-volt operation by connecting these two sections in parallel or series respectively. When required, the state of charge device is located in section 1-4. For batteries with State of Charge data output (PINs ending with “E”), the ground connection for the SOC device is located on the battery connector at pin 1. See table above for differences in performance among the three types.

[5] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): https://battery.army.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.

All battery types are subject to the transportation requirements of 49 CFR 173.185. M32271/1-12C, -12D, 12E, -32E, -42C, -42D and -42E batteries (solid cathode) contain more than 1 gram of lithium per cell and more than 2 grams per battery; M32271/1-22C, -22D and -22E (liquid cathode) batteries contain more than 0.5 grams per cell and 1 gram per battery. Check the appropriate Safety Data Sheet for actual lithium content. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

All battery types described by this specification sheet include a complete discharge device in order to render the battery non-reactive after use. If the battery is damaged, or if the device does not operate as indicated by an amber or yellow light, the battery is considered reactive waste. Any type, whether completely discharged or not, may be recycled after use. Universal waste rules, where applicable, apply only when recycling is the chosen disposal method.

The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using the Types I and II batteries covered by this specification sheet: Technical Bulletin 43-6135. Please contact the CECOM LCMC Directorate for Safety at usarmy.APG.cecom.mbx.amsel-sf-sec-cecom-lcmc--ds@army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using any battery described by this specification sheet will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy – SH
Air Force – 99
DLA – CC

Review activities:
Navy – AS
Air Force – 71

Preparing activity:
Army – CR
(Project Number 6135-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil or https://quicksearch.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:35Z
Check the source to verify that this is the current version before use.